The ElGamal Digital Signature Scheme

Setup Phase

- p is selected as a large prime so that the discrete logarithm problem (DLP) in Z_p is intractable.
- $\alpha \in Z_p^*$ is a primitive root of p.
- $\mathcal{P} = Z_p^*$.
- $\mathcal{A} = Z_p^* \times Z_{p-1}$.
- Key space $\mathcal{K} = \{(p, \alpha, a, \beta) : 1 < a < p, \beta = \alpha^a \pmod{p}\}$.
- The public key is (p, α, β).
- The private key is a.

The ElGamal Digital Signature Scheme (Continued...)

Signature Generation Phase

- Let $x \in Z_p^*$ be a message to be signed by the signer.
- Signer A selects a random number $k \in Z_{p-1}^*$.
- Signer A computes $\gamma = \alpha^k \pmod{p}$.
- Signer A computes $\delta = (x - a\gamma)k^{-1} \pmod{p-1}$.
- Signature on message x is $sig_K(x) = (\gamma, \delta)$.
- Signer sends the signed message $(x, (\gamma, \delta))$ to the verifier V.

The ElGamal Digital Signature Scheme (Continued...)

Signature Verification Phase

- Verifier V has $x, \gamma \in Z_p^*, \delta \in Z_{p-1}$, and public key (p, α, β).
- Verfier V computes $\beta^\gamma \pmod{p}$.
- Verfier V computes $\gamma^\delta \pmod{p}$.
- Verifier V checks the condition: $\beta^\gamma \gamma^\delta \equiv \alpha^x \pmod{p}$.
- If above holds, V accepts A's signature; otherwise, rejects it.

Correctness Proof of Signature Verification Phase

- We have,

$$\begin{aligned} \delta &= (x - a\gamma)k^{-1} \pmod{p-1} \\ k\delta &= x - a\gamma \pmod{p-1} \\ a\gamma + k\delta &= x \pmod{p-1} \end{aligned}$$

- Now,

$$\begin{aligned} \beta^{\gamma}\gamma^{\delta} &= (\alpha^{a} (\mathrm{mod}\ p))^{\gamma} (\alpha^{k} (\mathrm{mod}\ p))^{\delta} \pmod{p} \\ &= \alpha^{a\gamma + k\delta} \pmod{p} \\ &= \alpha^{x} \pmod{p} \end{aligned}$$

Example [ElGamal Digital Signature]

- Take a prime $p = 467$.
- Take a primitive root $\alpha = 2$.
- Choose randomly $a = 127$.
- Compute by repeated square-and-multiply algorithm,

$$\begin{aligned}\beta &= \alpha^a \pmod p \\ &= 2^{127} \pmod{467} \\ &= 132.\end{aligned}$$

- The public key is $(p, \alpha, \beta) = (467, 2, 132)$.
- The private key is $a = 127$.

Example [ElGamal Digital Signature] (Continued...)

- Suppose Alice wants to sign the message $x = 100$.
- Alice chooses the random number $k = 213 \in Z_p^*$. Note that $\gcd(k, p-1) = \gcd(213, 466) = 1$.
- Alice computes $k^{-1} \pmod{p-1} = 213^{-1} \pmod{466} = 431$, using the extended Euclid's GCD algorithm.
- Alice computes by repeated square-and-multiply algorithm,

$$\begin{aligned}
\gamma &= \alpha^k \pmod{p} \\
&= 2^{213} \pmod{467} \\
&= 29, \\
\delta &= (x - a\gamma)k^{-1} \pmod{p-1} \\
&= (100 - 127 \times 29) \times 431 \pmod{466} \\
&= 51.
\end{aligned}$$

The ElGamal Digital Signature Scheme (Continued...)

Example [ElGamal Digital Signature] (Continued...)

- Alice sends the signed message $(m, (\gamma, \delta)) = (100, (29, 51))$ to verifier, Bob.
- Bob verifies this signature by checking that $\beta^{\gamma}\gamma^{\delta} \equiv \alpha^{x} \pmod{p}$.
- Bob computes by repeated square-and-multiply algorithm,

$$\begin{aligned} \beta^{\gamma}\gamma^{\delta} &= 132^{29} \times 29^{51} \pmod{467} \\ &= 189, \\ \alpha^{x} &= 2^{100} \pmod{467} \\ &= 189. \end{aligned}$$

- Hence, the signature is verified.

Thank You